Arithmetic series

1 Describe the difference between a sequence and a series.

2 Find the sum of the first 15 terms of the following arithmetic series:

a $2 + 6 + 10 + \ldots$

b $-17 - 15 - 13 - \ldots$

c $35 + 31 + 27 + \ldots$

d $3 + 7 + 11 + \ldots$

e $5 - 2 - 9 - \ldots$

f $12 + 17 + 22 + \ldots$

3 Find the sum of the first 14 terms in the arithmetic series $13 + \ldots + \dfrac{13}{2}$.

4 Consider the arithmetic series $-9 - 5 - 1 + \ldots + 59$.

a Find n, the number of terms in the series.

b Find the sum of the series.

5 Consider the sum of the odd natural numbers: $1 + 3 + 5 + \ldots$ Show that the sum of the first n odd natural numbers is a perfect square.

6 Consider the arithmetic sequence $4, -1, -6, \ldots$

a Write a simplified expression for the sum of the first n terms.

b Find the sum of the progression from the 19th to the 27th term, inclusive.

7 For each of the following, find the sum of the first 10 terms of the arithmetic sequence:

a $u_1 = 6$ and $d = 3$

b $u_1 = -18$ and $d = 4$.

c $u_1 = 6$ and $u_{10} = 3.5$.

d $u_1 = -8$ and $u_{10} = -2.75$.

e $u_4 = 4$ and $u_7 = 13$.

f $u_6 = 2$ and $u_7 = 5$.

8 The first term of an arithmetic sequence is -5 and the sixth term is -45.

a Find the value of d.

b Find the sum of the first 14 terms.

9 Find the sum of the first 20 terms of the arithmetic series defined by $u_1 = \pi$ and $u_{20} = 20\pi$.

10 Find the sum of the first 40 terms of the following arithmetic sequences:

a $u_n = 6n$

b $u_n = 5 - 3n$

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11 In an arithmetic progression, the sum of the first n terms is denoted by S_n. If $S_{20} = 670$, and $S_{19} = 608$, find u_{20}.

12 The first term of an arithmetic sequence is 7, the common difference is 3 and the sum of the first n terms is 205. Find the value of n.

13 For each of the following:

i Find the common difference, d.

ii Find the first term, u_1.

iii Find the sum of the first 25 terms.

a $u_{10} = 34$ and $S_8 = 96$

b $u_9 = \frac{91}{5}$ and $u_{13} = \frac{99}{5}$.

c $u_8 = \frac{37}{2}$ and $u_{12} = \frac{41}{2}$

d $u_1 + u_3 = 0$ and $S_6 = 90$

e $S_8 = 16$ and $S_{14} = 196$

14 The sum of the first n terms of an arithmetic progression is $S_n = 3n^2 - 8n$.

a Find the value of:

i u_1 ii u_2 iii d

b Form a simplified expression for u_n, the nth term of the sequence.

15 Consider an arithmetic progression in which the terms are increasing. The first of three consecutive terms in the progression is y.

a Form an equation expressing d in terms of y, given that the sum of these three terms is -12.

b Form another equation relating y and d, given that the product of the three terms is 80.

c Find the value of y.

d State the value of the three consecutive terms.

Sigma notation

16 Consider the following and rewrite each series using summation notation:

a $\frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3}$

b $\frac{1}{3 \times 1} + \frac{1}{3 \times 2} + \frac{1}{3 \times 3} + \ldots + \frac{1}{3 \times 7}$

c $\frac{2}{\quad} + \frac{2}{\quad} + \frac{2}{\quad} + \ldots + \frac{2}{\quad}$

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17 Find the value of the following:

a $\sum_{k=3}^{5} \frac{1}{k}$ **b** $\sum_{r=1}^{4} \frac{1}{r+2}$ **c** $\sum_{k=1}^{5} (-1)^{k-1} k^2$ **d** $\sum_{r=-2}^{1} (3r^2 - 1)$

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a Write the following in expanded form:

i $\sum_{i=1}^{7} (i + i^3)$ **ii** $\sum_{i=1}^{7} i + \sum_{i=1}^{7} i^3$

b Do the above sums have the same value?

19 Show that:

$$\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$$

Technology

20 Use technology to find the sum of the first ten terms, S_{10}, of the following arithmetic series:

a $u_n = 4.2n + 8.81$ **b** $u_n = \sqrt{7}n + \sqrt{5}$

c $u_n = -\sqrt[3]{5}n + \sqrt{3}$

21 An arithmetic sequence has a 16th term of 267 and a 29th term of 449.

a Find the value of u_1.

b Find the value of d.

c Find the 34th term of the sequence.

22 The first three terms of an arithmetic sequence are $2x + 3$, $5x + 5$ and $8x + 7$.

a Find an expression for the common difference of the sequence.

b Find an expression for the sum S_{14}.

c If $S_{14} = 826$, use technology to calculate the value of x.

d Hence, state the general rule for u_n, the nth term of the arithmetic sequence.

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23 For each of the following, find the value of:

i u_1 ii d

a $u_{18} = 483$ and $S_{12} = 2484$

b $u_{17} = -307$ and $S_{12} = -534$

24 Use technology to find the term number, n, that will give the sum, S_n, for the following arithmetic sequences:

a $u_n = 39 + 12\,(n-1)$ where $S_n = 4914$

b $u_n = 23 - 11\,(n-1)$ where $S_n = -2035$

Applications

25 Find the sum of all integers between 20 and 50, inclusive, that are divisible by 6.

26 A worker at a factory is stacking cylindrical-shaped pipes in layers. Each layer contains one pipe less than the layer below it. There are 4 pipes in the topmost layer, 5 pipes in the next layer, and so on. There are n layers in the stack.

a Form an expression for the number of pipes in the bottom layer.

b Find the total number of pipes in the stack in terms of n.

27 Bart is learning to drive. His first lesson is 26 minutes long, and each subsequent lesson is 4 minutes longer than the lesson before.

a How long will his 15th lesson be?

b If Bart reaches 9.6 total hours of driving on his nth lesson, find the value of n.

28 For a sprint training exercise, a number of balls are placed in a straight line. The first ball is 4 metres from the start, and there is a 3 metre distance between each of the remaining consecutive balls. There are n balls placed out in the line in total.
Pauline must run from the start, collect the nearest ball and run back to the start, depositing the ball into a box. Then she must run back to collect the next ball and bring it back to the start. She continues this until all n balls have been collected.

a Write an expression for how far Pauline runs to collect and bring back the kth ball.

b Write an expression for how far she runs to collect and deposit all n balls.

c Pauline wants to runs 350 metres in total to collect and deposit all the balls. Solve for n, the number of balls that would need to be placed in a line.

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29 Katrina starts training for a $4.5\ \text{km}$ charity trail run by running every week for 28 weeks. She runs $2\ \text{km}$ of the course in the first week, and each week after that she runs 250 metres more than the previous week. She continues to increase the distance she runs until the week when she runs far enough to complete the course. Each week after that she completes the course without increase.

a How far does she run in the 11th week?

b What is the total distance that Katrina runs in 28 weeks? Round your answer to two decimal places.

30 A fishing trawler spends several days netting crabs. On the first day, it nets $550\ \text{kg}$ of crabs, on the next day it nets $538\ \text{kg}$. The amount netted continues to decrease by the same amount each day.

a How many kilograms are netted on the 11th day?

b What is the total weight netted in the first 11 days?

c The trawler returns to port when it has netted a total weight of $10\,400\ \text{kg}$. Find the number of days the trawler spent at sea.

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